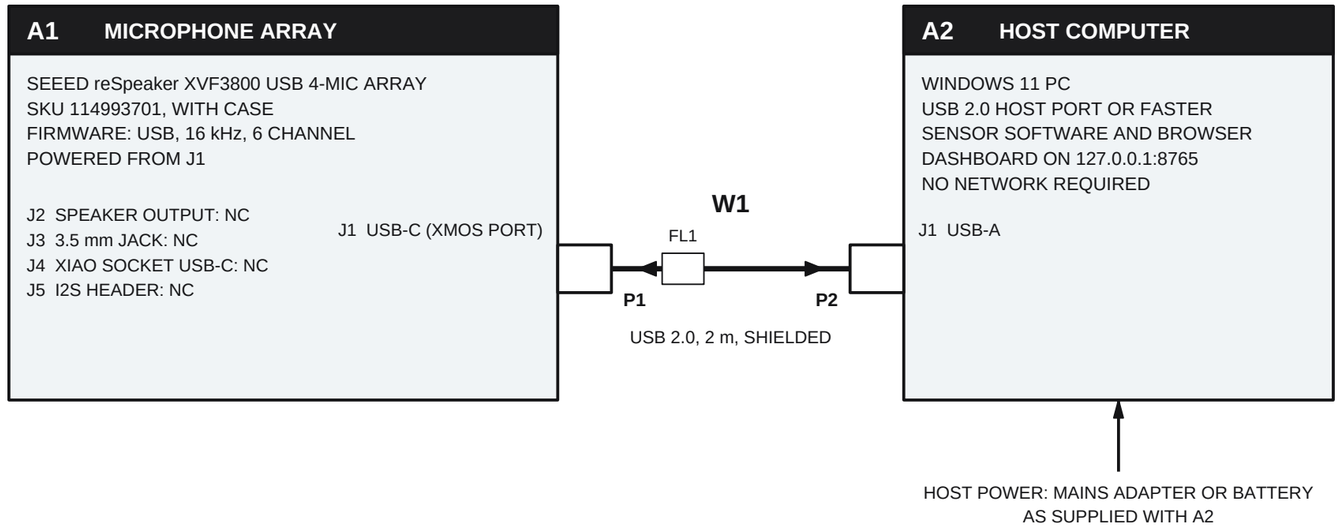
REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

INTERCONNECTION

ONE CABLE. NO OTHER ELECTRICAL CONNECTIONS



NOTES:

1. W1 CONNECTS A1-J1 DIRECTLY TO A HOST PORT. NO UNPOWERED HUB.
2. TOTAL USB CABLE LENGTH 3 m MAX WITHOUT AN ACTIVE EXTENSION.
3. A1 IS POWERED FROM J1 ONLY. SUPPLY CURRENT IS NOT PUBLISHED BY THE MANUFACTURER. MEASURE IT (TEST HB-3). LIMIT 500 mA, ONE USB 2.0 PORT.
4. A1-J2 TO J5 ARE NOT CONNECTED (NC). NO SOLDERING. NO MAINS WIRING.
5. FL1 IS FITTED ONLY IF INTERFERENCE IS SEEN ON THE AUDIO. PLACE IT WITHIN 50 mm OF P1 AND TIE IT IN PLACE.
6. PIN ASSIGNMENTS ARE THOSE OF A STANDARD USB 2.0 TYPE-C TO TYPE-A CABLE. W1 IS A PURCHASED PART AND IS NOT MODIFIED.
7. CONNECTOR DESIGNATORS ON A1 ARE ASSIGNED BY THIS DRAWING. THEY ARE NOT THE MANUFACTURER'S MARKINGS.
8. THE FOUR RAW MICROPHONES ARE CAPTURE CHANNELS 2 TO 5, COUNTING FROM 0, OF THE 6 CHANNEL 16 kHz USB AUDIO STREAM. SEE firmware/PROVISIONING.md.
9. PRELIMINARY. NOT VERIFIED ON HARDWARE.

WIRE LIST, CABLE W1

CONDUCTOR	W1-P1 PIN (USB-C PLUG)	W1-P2 PIN (USB-A)	SIGNAL
1	A4, A9, B4, B9	1	VBUS, +5 V
2	A7	2	D-
3	A6	3	D+
4	A1, A12, B1, B12	4	GND
NONE	A5 (CC)	NO CONNECTION	56 KILOHM PULL-UP TO VBUS INSIDE P1
SHIELD	SHELL	SHELL	CABLE SHIELD, BONDED AT BOTH ENDS

REFERENCE DESIGNATIONS

REF	DESCRIPTION	PART NUMBER
A1	MICROPHONE ARRAY WITH CASE	SEEED 114993701
A2	HOST COMPUTER	EXISTING WINDOWS 11 PC
W1	CABLE, USB-C TO USB-A, 2 m	STARTECH USB2AC2M
FL1	FERRITE, SNAP-ON (OPTIONAL)	FAIR-RITE 0431164281

UNLESS OTHERWISE SPECIFIED
DIAGRAM. NOT TO SCALE
NO DIMENSIONS ON THIS SHEET

THIRD ANGLE PROJECTION

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL: NOT APPLICABLE

FINISH: NOT APPLICABLE

STATUS: PRELIMINARY

S&L MILESTONE SOLUTIONS

TITLE
INTERCONNECTION DIAGRAM
PASSIVE ACOUSTIC SENSOR NODE

SIZE	DWG NO	REV
B	SLM-PAS-E01	A

SCALE	PROJECT	SHEET
NONE	PAS NODE	1 OF 1